

ISW Void Sign-Flip Observational Pipeline

From EFC Theory to Data Confrontation

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Abstract. This document specifies the complete data pipeline required to observationally test the EFC void ISW sign-flip prediction. The pipeline takes the existing theoretical framework (v2.1) and confronts it with publicly available void catalogs, Planck CMB temperature and lensing maps, and large-scale structure survey data. The deliverable is a measured amplitude ratio $A_{\text{ISW}}(\delta)$ as a function of void depth, which either confirms or falsifies the predicted sign reversal at $\delta_c \approx -0.8$. We specify every data source, processing step, stacking methodology, statistical test, and kill criterion. The pipeline is designed to run on a single workstation with ~ 32 GB RAM.

1 Motivation and EFC Prediction

1.1 The ISW Anomaly in Λ CDM

The ISW amplitude from stacked supervoids has been measured at $A_{\text{ISW}} \approx 5.2 \pm 1.6$ using BOSS DR12 and DES Y3 data in the redshift range $0.2 < z < 0.9$, a value that substantially exceeds the Λ CDM expectation of $A_{\text{ISW}} = 1$. This excess has persisted across multiple independent analyses (Granett et al. 2008; Kovács et al. 2017, 2019, 2022) and remains unexplained within the standard model. The prevailing view is that no modification of Λ CDM parameters, within existing constraints, can accommodate the signal.

1.2 EFC Prediction: Sign-Flip in Deep Voids

In Energy-Flow Cosmology, the gravitational coupling μ depends on local density via $\mu(\rho) \propto \sqrt{\rho/\rho_{\text{crit}}}$. This produces a modified ISW decomposition:

$$\frac{d\Phi}{dt} = \underbrace{\mu \cdot \frac{d\delta}{dt}}_{\text{Linear ISW (cold)}} + \underbrace{\delta \cdot \frac{d\mu}{dt}}_{\text{Rees-Sciama (hot)}} \quad (1)$$

The second term is the EFC-specific Rees–Sciama (RS) contribution. In voids ($\delta < 0$), the density decreases over time ($d\rho/dt < 0$), so $d\mu/dt < 0$. Since $\delta < 0$ and $d\mu/dt < 0$, the RS term is *positive* — it produces a hot spot that opposes the standard cold ISW signal.

The total amplitude ratio relative to Λ CDM is:

$$A_{\text{total}}(\delta) = \frac{\Delta T_{\text{EFC}}}{\Delta T_{\Lambda\text{CDM}}} = 1 + \frac{\text{RS}}{\text{ISW}_{\text{lin}}} \quad (2)$$

Three quantitative predictions emerge (from the v2.1 theory paper):

Table 1: EFC Void ISW Predictions (frozen parameters)

ID	Prediction	Threshold	Kill criterion
P1	Depth turnover: A_{ISW} peaks then flips	$\delta_c \approx -0.8$	No sign-flip at any δ
P2	Scale: larger R_v amplifies RS	$R_v > 100 h^{-1}\text{Mpc}$	No scale trend
P3	Redshift: stronger at lower z	$z < 0.5$ strongest	No z -evolution

1.3 Why This Test Is Decisive

The sign-flip is a *qualitative* prediction: Λ CDM predicts $A_{\text{ISW}} > 0$ (cold spots in voids) at all void depths. EFC predicts A_{ISW} turns *positive* (hot spots) for $\delta \lesssim -0.8$. No parameter tuning in Λ CDM can produce this behavior. A confirmed sign-flip at the predicted depth would be the single strongest discriminator between EFC and Λ CDM.

2 Data Sources

2.1 Void Catalogs

Table 2: Available void catalogs for the pipeline

Catalog	Survey	N_{voids}	z_{max}	Access
DESIVAST DR1	DESI BGS	~ 1500	0.24	Public
Sutter+ 2014	SDSS DR7	~ 1000	0.44	Public
BOSS DR12 supervoids	SDSS-III	~ 100	0.70	Kovács+ 2018
DES Y3 supervoids	DES	~ 200	0.90	Kovács+ 2022
eBOSS DR16 QSO	SDSS-IV	~ 800	2.20	Kovács+ 2022
Quaia QSO voids	Gaia/WISE	~ 500	2.20	Sartori+ 2025

Primary dataset: DESIVAST DR1 (VoidFinder algorithm), publicly available at <https://data.desi.lbl.gov/public/dr1/vac/dr1/desivast/>. This catalog has the best depth characterization (δ per void from Voronoi density reconstruction) and is volume-limited, eliminating Malmquist bias concerns.

Secondary datasets: BOSS/DES supervoid catalogs provide access to deeper voids ($\delta < -0.5$) at higher redshift, where the EFC effect should be measurable but the sign-flip threshold harder to reach.

2.2 CMB Maps

- **Temperature:** Planck PR4 SMICA/SEVEM/NILC/Commander maps at $N_{\text{side}} = 2048$
 - **Lensing convergence:** Planck 2018 κ map (for cross-check: lensing should show *no* sign-flip)
 - **Masks:** Planck UT78 common confidence mask, point source masks
- All maps available from the Planck Legacy Archive (pla.esac.esa.int).

2.3 Simulations for Null Testing

- **MICE:** Grand Challenge simulation, provides mock CMB lensing for calibration
- **Jubilee:** ISW simulation used as Λ CDM baseline for A_{ISW} expectation
- **AbacusSummit:** DESI-specific mocks for DESIVAST null tests

3 Pipeline Architecture

The pipeline consists of six sequential stages. Each stage has a defined input, output, and validation check.

3.1 Stage 1: Void Catalog Ingestion

Input: Raw DESIVAST FITS files (VoidFinder NGC + SGC).

Processing:

1. Read void centers (RA, Dec, z , R_{eff})
2. Compute void density contrast δ from the catalog's Voronoi density field
3. Apply quality cuts: remove edge voids ($> 20\%$ surface on survey boundary), $R_{\text{eff}} > 10 h^{-1}\text{Mpc}$
4. Bin voids by δ in 5 bins: $[-0.3, -0.5)$, $[-0.5, -0.6)$, $[-0.6, -0.7)$, $[-0.7, -0.8)$, $[-0.8, -1.0)$
5. Bin voids by R_{eff} : small ($< 30 h^{-1}\text{Mpc}$), medium (30–80), large (> 80)
6. Bin voids by z : low (< 0.10), mid (0.10–0.18), high (0.18–0.24)

Output: Structured catalog with per-void δ , R_{eff} , z , bin assignments.

Validation: Reproduce Table 3 of Rincon et al. (2025) void size distribution.

3.2 Stage 2: CMB Map Preparation

Input: Planck PR4 maps + masks.

Processing:

1. Read SMICA temperature map at $N_{\text{side}} = 2048$
2. Apply UT78 mask $\times$ point-source mask
3. Downgrade to $N_{\text{side}} = 512$ (sufficient for ISW scales $> 1^\circ$)
4. Subtract monopole and dipole from unmasked pixels
5. Optional: apply Gaussian smoothing FWHM = $30'$ to suppress small-scale noise
6. Repeat for SEVEM, NILC, Commander (systematic cross-check)

Output: Four cleaned CMB temperature maps at $N_{\text{side}} = 512$.

Validation: Power spectrum C_ℓ matches Planck published values at $\ell < 100$.

3.3 Stage 3: Stacking

This is the core measurement. We use the Compensated Top-Hat (CTH) filter following Kovács (2018).

For each void:

1. Extract a $10^\circ \times 10^\circ$ CMB temperature patch centered on (RA, Dec)
2. Rescale angular coordinates to units of $R_v/D_A(z)$ (physical void radius / angular diameter distance)
3. Apply CTH filter with radius R_{filt} :

$$W(\theta) = \begin{cases} +1/\pi R_{\text{filt}}^2 & \theta < R_{\text{filt}} \\ -1/\pi(4R_{\text{filt}}^2 - R_{\text{filt}}^2) & R_{\text{filt}} < \theta < 2R_{\text{filt}} \\ 0 & \theta > 2R_{\text{filt}} \end{cases} \quad (3)$$

4. Record filtered temperature ΔT_{filt} for this void

Stacking:

1. Average ΔT_{filt} over all voids in each $(\delta, R_{\text{eff}}, z)$ bin
2. Estimate uncertainty via bootstrap (10 000 re-samplings) and random-catalog comparison
3. Compute $A_{\text{ISW}}(\delta) = \langle \Delta T_{\text{obs}} \rangle / \langle \Delta T_{\Lambda\text{CDM}} \rangle$ per bin

Output: $A_{\text{ISW}}(\delta)$ with 1σ errors for each bin.

Normalization robustness: All results are reported both as A_{ISW} (normalized to ΛCDM) and as raw $\langle\Delta T\rangle$ in μK . Conclusions regarding sign-flip rely only on the *sign* of $\langle\Delta T\rangle$, not on the normalization to ΛCDM . This removes dependence on the ΛCDM denominator, which is itself anomalous ($A_{\text{ISW}}^{\text{obs}} \approx 5$).

3.4 Stage 4: EFC Model Prediction

Input: Void bins from Stage 1, EFC theoretical code from `void_isw.py` (v2.1).

Processing:

1. For each δ -bin, compute EFC-predicted $A_{\text{total}}(\delta)$ using Eq. (2)
2. Propagate uncertainty in δ within each bin
3. Generate predicted $A_{\text{ISW}}(\delta)$ curve with error band

Output: EFC prediction curve for direct overlay on Stage 3 data.

3.5 Stage 5: Statistical Comparison

Tests:

1. χ^2 **test:** Compute χ_{EFC}^2 and $\chi_{\Lambda\text{CDM}}^2$ for $A_{\text{ISW}}(\delta)$ data vs. each model
2. $\Delta\chi^2$: Report $\Delta\chi^2 = \chi_{\Lambda\text{CDM}}^2 - \chi_{\text{EFC}}^2$
3. **Sign-flip test:** In the deepest δ -bin, test $H_0: A_{\text{ISW}} > 0$ (cold) vs. $H_1: A_{\text{ISW}} < 0$ (hot). Report p -value.
4. **Trend test:** Spearman rank correlation between $|\delta|$ and A_{ISW} for monotonic depth dependence
5. **Lensing cross-check:** Repeat stacking on κ map. EFC predicts *no* sign-flip in lensing (lensing probes Σ , not Φ). Anomalous lensing sign-flip would indicate systematics.

Output: $\Delta\chi^2$, p -values, trend statistics, lensing null check.

3.6 Stage 6: Verdict

Apply the pre-registered kill criteria:

Table 3: Kill criteria and verdicts

Observation	Verdict for EFC	Verdict for ΛCDM
Sign-flip at $ \delta > 0.8$ with $p < 0.05$	Strong support	Falsified at this depth
A_{ISW} monotonically rising with $ \delta $	Consistent (partial)	Anomalous ($A > 1$)
$A_{\text{ISW}} \approx 1$ at all δ	Falsified (no RS)	Confirmed
Sign-flip in lensing κ too	Both models fail	Systematic error

4 Implementation

4.1 Software Dependencies

Table 4: Required Python packages

Package	Version	Purpose
healpy	≥ 1.16	HEALPix map operations
astropy	≥ 6.0	FITS I/O, coordinates, cosmology
numpy	≥ 1.26	Array operations
scipy	≥ 1.12	Statistics, interpolation
matplotlib	≥ 3.8	Plotting
pixell	≥ 0.22	Flat-sky patch extraction
camb	≥ 1.5	Λ CDM theory predictions

4.2 Core Pipeline Code

The pipeline is implemented as a single Python module. Below is the complete skeleton with all critical functions.

Listing 1: Pipeline Stage 1: Void catalog ingestion

```
import numpy as np
from astropy.io import fits
from astropy.table import Table
from astropy.cosmology import Planck18 as cosmo

def load_desivast(ngc_path, sgc_path):
    """Load DESIVAST VoidFinder catalogs."""
    ngc = Table.read(ngc_path)
    sgc = Table.read(sgc_path)
    cat = vstack([ngc, sgc])
    # Quality cuts
    mask = (cat['EDGE_AREA'] / cat['TOTAL_AREA'] < 0.2)
    mask &= (cat['EFFECTIVE_RADIUS'] > 10.0)
    return cat[mask]

def compute_delta(cat):
    """Estimate void density contrast from catalog."""
    # DESIVAST provides volume; delta from
    # tracer density relative to mean
    rho_mean = len(cat) / cat['VOLUME'].sum()
    cat['delta'] = (cat['VOID_DENSITY'] / rho_mean) - 1
    return cat

def bin_voids(cat):
    """Bin voids by delta, Reff, redshift."""
    delta_edges = [-1.0, -0.8, -0.7, -0.6, -0.5, -0.3]
    R_edges = [10, 30, 80, 300]
    z_edges = [0.0, 0.10, 0.18, 0.24]

    cat['delta_bin'] = np.digitize(cat['delta'], delta_edges)
    cat['R_bin'] = np.digitize(
        cat['EFFECTIVE_RADIUS'], R_edges)
```

```
cat['z_bin'] = np.digitize(cat['z'], z_edges)
return cat
```

Listing 2: Pipeline Stage 3: CMB stacking with CTH filter

```
import healpy as hp

def prepare_cmb(map_path, mask_path, nside_out=512):
    """Load and clean Planck CMB temperature map."""
    T = hp.read_map(map_path, field=0)
    mask = hp.read_map(mask_path)
    T[mask < 0.5] = hp.UNSEEN

    # Downgrade
    T_low = hp.ud_grade(T, nside_out)
    mask_low = hp.ud_grade(mask, nside_out)

    # Remove monopole + dipole
    T_clean = hp.remove_dipole(
        T_low, gal_cut=20, fitval=True)[0]
    return T_clean, mask_low

def cth_filter(patch, R_filt_pix):
    """Compensated Top-Hat filter on a 2D patch."""
    ny, nx = patch.shape
    cy, cx = ny // 2, nx // 2
    y, x = np.ogrid[:ny, :nx]
    r = np.sqrt((x - cx)**2 + (y - cy)**2)

    inner = r < R_filt_pix
    outer = (r >= R_filt_pix) & (r < 2 * R_filt_pix)

    A_in = np.sum(inner)
    A_out = np.sum(outer)

    if A_in == 0 or A_out == 0:
        return np.nan

    T_cth = (np.nanmean(patch[inner])
             - np.nanmean(patch[outer]))
    return T_cth

def stack_voids(cat, T_map, nside, R_filt_deg=2.0):
    """Stack CMB patches on void positions."""
    results = []
    for void in cat:
        ra, dec, z = void['RA'], void['DEC'], void['z']
        R_v = void['EFFECTIVE_RADIUS'] #  $h^{-1}$  Mpc
        D_A = cosmo.angular_diameter_distance(z).value
        theta_v = np.degrees(R_v / D_A) # degrees

        # Extract patch
        vec = hp.ang2vec(ra, dec, lonlat=True)
        ipix = hp.query_disc(nside, vec,
                             np.radians(5 * theta_v))

        # Project to flat patch
```

```

    patch = extract_flat_patch(
        T_map, nside, ra, dec,
        size_deg=10*theta_v)

    R_filt_pix = R_filt_deg / theta_v * (
        patch.shape[0] / (10 * theta_v))
    dT = cth_filter(patch, R_filt_pix)

    results.append({
        'delta': void['delta'],
        'R_eff': R_v,
        'z': z,
        'dT': dT,
        'theta_v': theta_v
    })
return Table(results)

```

Listing 3: Pipeline Stage 4–5: EFC prediction and comparison

```

import sys
sys.path.insert(0, 'void_isw_signflip_v2.1_final/src')
from void_isw import (ISWDecomposition, AmplitudeRatio,
                      Predictions)

def efc_prediction_curve(delta_bins):
    """Compute EFC A_total for each delta bin."""
    decomp = ISWDecomposition()
    predictions = []
    for d_lo, d_hi in delta_bins:
        d_mid = 0.5 * (d_lo + d_hi)
        result = decomp.compute(delta=d_mid, z=0.15)
        predictions.append({
            'delta_mid': d_mid,
            'A_total': result['A_total'],
            'sign_flip': result['sign_flip'],
            'isw_lin': result['isw_linear'],
            'rees_sciamma': result['rees_sciamma']
        })
    return Table(predictions)

def compute_chi2(data, model):
    """Chi-squared comparison."""
    residuals = data['A_ISW'] - model['A_total']
    chi2 = np.sum(residuals**2 / data['A_ISW_err']**2)
    return chi2

def sign_flip_test(data, deepest_bin_idx=-1):
    """Test H0: A_ISW > 0 vs H1: A_ISW < 0."""
    A = data['A_ISW'][deepest_bin_idx]
    sigma = data['A_ISW_err'][deepest_bin_idx]
    # One-sided p-value for A < 0
    from scipy.stats import norm
    p_value = norm.cdf(A / sigma)
    return A, sigma, p_value

def lcdm_prediction(delta_bins):
    """LCDM predicts A_ISW = 1 at all depths."""

```



```
return Table({
    'delta_mid': [0.5*(lo+hi) for lo,hi in delta_bins],
    'A_total': np.ones(len(delta_bins))
})
```

5 Execution Timeline

Table 5: Execution timeline and milestones

Week	Stage	Milestone	Deliverable
1	Data acquisition	Download all catalogs + maps	Raw data on disk
2	Stage 1–2	Void catalog + CMB ready	Binned void catalog, cleaned maps
3	Stage 3	Stacking complete	$A_{\text{ISW}}(\delta)$ per bin
4	Stage 4–5	Model comparison	$\Delta\chi^2$, p -values
5	Stage 6	Verdict + paper draft	Validation ledger entry

5.1 Resource Requirements

The pipeline is designed to be computationally modest:

- **Storage:** ~ 5 GB for Planck maps + ~ 200 MB for void catalogs
- **RAM:** ~ 16 GB (HEALPix maps at $N_{\text{side}} = 2048$; downgraded for stacking)
- **CPU:** Stacking 1500 voids with bootstrap: ~ 2 hours on 8 cores
- **No GPU required**

5.2 Running on the Symbiose Machine

The pipeline is designed to run on the Symbiose research server. The existing `void_isw_signflip_v2.1.final/` package provides the EFC theory predictions. Stage 1 data download can be automated via:

Listing 4: Data download commands

```
# DESIVAST void catalog
wget -r -np -nH --cut-dirs=5 \
  https://data.desi.lbl.gov/public/dr1/vac/dr1/desivast/

# Planck PR4 SMICA map
wget https://pla.esa.int/pla/aio/product-action\
?MAP.MAP_ID=COM_CMB_IQU-smica_2048_R3.00_full.fits

# Planck lensing convergence
wget https://pla.esa.int/pla/aio/product-action\
?MAP.MAP_ID=COM_Lensing_4096_R3.00_TT_kappa.fits
```

6 Systematic Controls

6.1 Known Systematics and Mitigations

Table 6: Systematic effects and control strategies

Systematic	Risk	Mitigation
CMB foreground residuals	Dust/synchrotron leak into temperature	Stack using all 4 component-separation maps; require consistency
Void definition dependence	Result depends on void-finder	Compare VoidFinder vs. VIDE vs. REVOLVER from DESIVAST
A posteriori filter tuning	Inflated significance	Fix $R_{\text{filt}} = R_v$ before unblinding; no α -rescaling
Photo- z smearing	Void depth diluted	DESI spectroscopic z eliminates this
Void overlap/alignment	Correlated stacking noise	Jackknife by survey region

6.2 Null Tests

1. **Random catalog stacking:** Stack on $10\times$ the number of random positions; should be consistent with zero.
2. **Rotated patches:** Rotate each CMB patch by 90° before stacking; should destroy signal.
3. **Lensing convergence:** Stack κ map on same void positions. EFC predicts NO sign-flip in lensing (it only affects $\dot{\Phi}$, not Φ itself). A sign-flip in κ would indicate an error.
4. **Cross-map consistency:** SMICA, SEVEM, NILC, Commander must give consistent A_{ISW} within 1σ .
5. **Hemisphere split:** NGC vs. SGC must be consistent.

6.3 Blinding Protocol

To prevent a posteriori bias:

1. Fix all analysis choices (binning, filter, cuts) on mock catalogs from AbacusSummit
2. Freeze the EFC prediction curve before looking at data
3. Unblind only after null tests pass
4. Report $A_{\text{ISW}}(\delta)$ for all bins simultaneously, not selectively

7 Connection to Existing EFC Infrastructure

7.1 Integration with the Living Validation Ledger

This pipeline maps to Hull 6 in the gap analysis. Upon completion, the result enters the ledger as a formal validation test:

Table 7: Ledger entry specification

Field	Value
Test ID	ISW-VOID-SIGNFLIP-OBS-v1
Category	physics_test
Track	Spor 1 — Galactic/Cosmological
Prediction	Sign-flip at $\delta_c \approx -0.8$
Data source	DESIVAST DR1 + Planck PR4
Kill criterion	No sign-flip at any δ
Status	planned_pipeline

7.2 Existing Code Assets

The following files in the Symbiose repository are directly used:

- `void_isw_signflip_v2.1_final/src/void_isw.py` — all EFC theory calculations
- `void_isw_signflip_v2.1_final/data/void_isw_data.json` — reference data points
- `void_isw_signflip_v2.1_final/schema.json` — output validation schema
- `void_isw_signflip_v2.1_final/examples/demo_void_isw.py` — demonstration code

7.3 New Code Required

Table 8: New modules to be written

Module	Lines (est.)	Description
<code>ingest_desivast.py</code>	~150	FITS reader, quality cuts, binning
<code>cmb_prep.py</code>	~100	Map loading, masking, downgrading
<code>stacker.py</code>	~250	CTH filter, patch extraction, bootstrap
<code>compare.py</code>	~120	χ^2 , sign-flip test, trend test
<code>plot_result.py</code>	~80	$A_{\text{ISW}}(\delta)$ figure
Total	~700	

8 Expected Outcomes

8.1 If EFC Is Correct

The stacked ISW signal in the deepest void bin ($\delta < -0.8$) should show a *positive* temperature anomaly (hot spot) instead of the Λ CDM-predicted cold spot. The $A_{\text{ISW}}(\delta)$ curve should show:

- $A_{\text{ISW}} \approx 1\text{--}3$ for shallow voids ($\delta \approx -0.3$)
- A_{ISW} rising with $|\delta|$ due to growing RS term
- Turnover and sign-flip near $\delta_c \approx -0.8$
- Negative A_{ISW} (hot spot) for $\delta < \delta_c$

This would explain the observed $A_{\text{ISW}} \approx 5.2 \pm 1.6$ excess as the RS enhancement in EFC, and predict further amplification for deeper voids.

8.2 If EFC Is Wrong

$A_{\text{ISW}}(\delta) \approx 1$ at all depths (or follows Λ CDM noise). No sign-flip, no depth trend beyond statistical scatter. The pipeline kills the ISW prediction and the Validation Ledger records a formal failure.

8.3 Intermediate Cases

- **Enhanced A_{ISW} but no sign-flip:** RS term present but weaker than predicted. Constrains $\mu(\rho)$ slope. Partial validation.
- **Too few deep voids:** DESIVAST ($z < 0.24$) may not contain voids with $\delta < -0.8$. Fall back to BOSS/DES supervoid catalogs. Record as “insufficient data, not falsified.”
- **Lensing shows anomaly too:** Points to systematic error in void density estimation, not a physical signal. Suspend verdict pending investigation.

9 RCMB Compliance Layer

This section applies the Regime-Bound Measurement in Complex Systems (RCMB) framework to the ISW pipeline. RCMB does not replace the physics test — it audits the *measurement validity* of the test. The structural chain Regime $\rightarrow$ Proxy + Placement $\rightarrow$ Episenter $\rightarrow$ Compression $\rightarrow$ Output is applied to every stage.

9.1 Regime Declaration

The pipeline operates across three regimes simultaneously. RCMB requires each to be explicitly declared and matched.

Table 9: Regime declaration for ISW pipeline

Regime	Nature	Quantity	Mismatch risk
EFC theory	Local ρ at void center	$\mu(\rho), \delta_c$	Prediction is local; data is averaged
Void cata-log	Tracer galaxies, volume metric	$\delta_{\text{VoidFinder}}$	$\delta_{\text{tracer}} \neq \delta_{\text{matter}}$ (bias b)
CMB signal	Line-of-sight integral	$\Delta T \propto \int \dot{\Phi} d\eta$	Not local to void; accumulates LoS

Consequence: The sign-flip threshold $\delta_c \approx -0.8$ is defined in *physical* density space. The observable threshold δ_c^{obs} depends on tracer bias, void-finder smoothing, and line-of-sight mixing:

$$\delta_c^{\text{obs}} = f(b, \sigma_{\text{smooth}}, \Delta z_{\text{LoS}}) \neq \delta_c^{\text{theory}} \quad (4)$$

The pipeline therefore tests: “does sign-flip exist in observable space?” — not “is $\delta_c = -0.8$ exactly?”

Uncertainty propagation: Each δ -bin represents a *distribution* of true densities due to tracer bias and void-finder smoothing. All comparisons to EFC predictions are therefore interpreted as convolutions over this distribution, not point evaluations at bin centers. The EFC prediction curve $A_{\text{total}}(\delta)$ is convolved with the per-bin δ -scatter before computing χ^2 .

9.2 Proxy Chain Audit

RCMB Thesis 2 states: “Proxy creates both visibility and blindness.” The full proxy chain from physical target to measured output is:

Each blindness must be addressed by a specific control test (see Section 9.6).

9.3 Placement Declaration

RCMB defines three placement dimensions. All three contain assumptions in this pipeline:

Physical placement: Observer at $z = 0$, observing projected 2D imprints of 3D voids. Depth information along line-of-sight is lost in projection.

Instrument placement: Planck satellite, multi-frequency, component-separated maps. Different pipelines (SMICA/SEVEM/NILC/Commander) constitute different instrument placements. Consistency across them tests instrument-placement stability.

Interpretive placement: This is the critical one. The entire pipeline assumes Λ CDM geometry: $D_A(z)$ from Planck18, void identification uses cosmological distances, $\Delta T_{\Lambda\text{CDM}}$ from Λ CDM simulations. EFC is tested *through* a Λ CDM interpretive lens.

Table 10: Complete proxy chain with declared blindness

Step	Transformation	Declared blindness
0	$d\Phi/dt$ (target)	Not directly observable
1	$\rightarrow \Delta T_{\text{ISW}}$ via LoS integral	Accumulates signal from all structures along LoS, not just target void
2	$\rightarrow \Delta T_{\text{obs}}$ via component separation	Foreground residuals from dust, synchrotron, tSZ
3	$\rightarrow \Delta T_{\text{filt}}$ via CTH filter	Assumes circular symmetry; voids are elongated ($R_{\parallel}/R_{\perp} \approx 2.6$)
4	$\rightarrow \langle \Delta T \rangle_{\text{bin}}$ via stacking	Per-void variation compressed to mean; bimodality invisible
5	$\rightarrow A_{\text{ISW}}$ via Λ CDM normalization	Denominator itself anomalous (observed $A \approx 5$)

Mitigation: Run sensitivity test with $D_A \pm 10\%$; report whether sign-flip threshold shifts. If EFC background recovery (Level 3) becomes available, rerun with EFC-consistent $D_A(z)$.

9.4 Episenter-Shift Test

RCMB Thesis 4: “Episenter is produced, not given.” The pipeline’s default episenter is void depth δ . All results are organized by δ -bins. But δ is itself a derived quantity from the void-finder.

Test: Recompute A_{ISW} binned by three alternative epistemic centers:

1. R_{eff} (void size) — tests scale dependence
2. z (redshift) — tests evolution dependence
3. Ellipticity $e = 1 - R_{\perp}/R_{\parallel}$ — tests morphology dependence

RCMB prediction: If δ is a physically meaningful episenter, then episenter-shift should produce *structured* redistribution of residuals (not random scatter). If shifting episenter from $\delta \rightarrow R_{\text{eff}}$ makes the trend *disappear*, then δ was not the controlling variable — the void-finder was generating the structure. If multiple episenters (δ , R_v , z) show equally strong structured trends, the signal is not uniquely attributable to void depth and is considered confounded.

9.5 Compression Audit

RCMB Thesis 5: “Compression loss has predictable footprint.”

The stacking operation $\{\Delta T_i\} \rightarrow \langle \Delta T \rangle$ is a severe compression. If EFC is correct and sign-flip occurs at $\delta_c \approx -0.8$, then for voids near the threshold, some will have flipped and some will not (due to scatter in true δ vs. measured δ).

Consequence: The ΔT distribution in the deepest bin should be:

- **Broader** than shallower bins (increased variance)
- **Non-Gaussian:** positive skewness or bimodality
- The **mean may be near zero** even though the physics is real (cancellation of flipped and unflipped voids)

Critical insight: The sign-flip will first appear as *increased variance and asymmetry*, not as a mean shift. A pipeline that only reports $\langle \Delta T \rangle$ can miss the signal entirely.

9.6 RCMB Control Tests

Four additional tests are required for RCMB compliance:

Table 11: RCMB control tests

ID	Test	What it checks
R1	Distribution analysis per bin	Compression audit: skewness, kurtosis, bimodality in deepest bin
R2	Episenter-shift ($\delta \rightarrow R_v \rightarrow z \rightarrow e$)	Episenter stability
R3	Multi-void-finder comparison	Proxy validity: VoidFinder vs. VIDE vs. REVOLVER
R4	LoS: z -shuffle + angular randomization	Placement: is signal local or projected?

9.7 Decision Hierarchy

The RCMB compliance layer has logical precedence over the physics verdict.

- If RCMB criteria (R1–R4) fail: the measurement is invalid and no physics conclusion is permitted.
- If RCMB criteria pass: Stage 6 verdict is considered valid.

Interpretation: RCMB determines whether the pipeline measures the intended quantity. Stage 6 determines what that measurement implies physically. These are orthogonal layers: RCMB audits the measurement; physics interprets the result. A valid kill decision requires both layers to be satisfied.

LoS decision rule: Both z -shuffle and angular-randomization tests must reject the null ($S/N > 2$) for the signal to be considered local. Failure of either test indicates projection contamination and invalidates the measurement. No physical interpretation is permitted until locality is established.

9.8 Minimal Detection Condition

A sign-flip candidate requires at least one of:

- Deepest bin mean consistent with $A_{\text{ISW}} < 0$ at $\geq 2\sigma$, **or**
- RCMB signature present: increased variance and non-Gaussianity (skewness $\neq 0$ at 2σ or bimodality coefficient > 0.555) in deepest bin relative to shallower bins.

If neither condition is met, no evidence for sign-flip is claimed. Conversely, a null mean ($\langle \Delta T \rangle \approx 0$) does *not* falsify the model if higher-order moments indicate structure — the compression layer may be cancelling flipped and unflipped voids. The primary detection channel for sign-flip is expected to be variance and higher-order moments, not the mean signal.

Higher-moment significance: Significance of non-Gaussianity is assessed via bootstrap resampling (10 000 iterations) within each bin. For skewness and kurtosis, a deviation from zero at $\geq 2\sigma$ (bootstrap confidence interval) is required. For bimodality, the coefficient threshold > 0.555 must be exceeded with stability under resampling (i.e., $> 95\%$ of bootstrap samples must exceed the threshold).

Bin stability test: All results are re-evaluated under small perturbations of bin edges (± 0.05 in δ). A genuine signal must remain stable under these perturbations. If sign-flip appears or disappears under bin-edge shifts, it is flagged as a binning artifact.

Definition of σ : All quoted σ values refer to bootstrap-estimated standard deviations from 10 000 resamples within each bin, unless otherwise stated.

Multiple testing control: Primary inference is restricted to the pre-registered deepest δ -bin. All additional tests (epi-center-shift, higher moments, bin stability) are treated as diagnostic and not as independent discovery channels. No look-elsewhere correction is applied to the primary test; diagnostics are reported at face value.

10 RCMB Implementation Code

The following ~200 lines extend the base pipeline with RCMB compliance.

Listing 5: RCMB Module 1: Proxy Chain Audit

```
"""rcmb_proxy.py -- Proxy chain documentation."""
import json, datetime

def proxy_chain_report(void, theta_v, cosmology="Planck18"):
    """Generate RCMB proxy chain for a single void.

    Every measurement step is logged with its
    transformation, assumption, and declared blindness.
    """
    chain = [
        {"step": 0, "name": "physical_target",
         "quantity": "dPhi/dt at void center",
         "status": "not directly observable"},
        {"step": 1, "name": "LoS_integral",
         "quantity": f"DeltaT_ISW (LoS through z={void['z']:.3f})",
         "assumption": "void dominates LoS signal",
         "blindness": "foreground/background structures"},
        {"step": 2, "name": "component_separation",
         "quantity": "DeltaT_obs (SMICA/SEVEM/...)",
         "assumption": "foregrounds fully removed",
         "blindness": "residual dust/synchrotron"},
        {"step": 3, "name": "CTH_filter",
         "quantity": f"DeltaT_filt (R_filt={theta_v:.2f} deg)",
         "assumption": "circular symmetry",
         "blindness": "void elongation (R_par/R_perp~2.6)"},
        {"step": 4, "name": "stacking",
         "quantity": "<DeltaT> per bin",
         "assumption": "single underlying distribution",
         "blindness": "per-void variance, bimodality"},
        {"step": 5, "name": "LCDM_normalization",
         "quantity": "A_ISW = DeltaT_obs / DeltaT_LCDM",
         "assumption": f"D_A from {cosmology}",
         "blindness": "LCDM geometry assumed for EFC test"}
    ]
    return {
        "void_id": void.get('VOID_ID', 'unknown'),
        "proxy_chain": chain,
        "timestamp": datetime.datetime.utcnow().isoformat(),
        "rcmb_version": "1.0"
    }
```

Listing 6: RCMB Module 2: Distribution Analysis (Compression Audit)

```
"""rcmb_distribution.py -- Compression audit."""
import numpy as np
from scipy import stats

def distribution_analysis(dT_values, bin_label=""):
    """Full distribution stats per delta-bin.

    RCMB requires reporting beyond the mean:
    variance, skewness, kurtosis, and bimodality test.
    """
```

```

"""
dT = np.array(dT_values)
dT = dT[np.isfinite(dT)]
n = len(dT)
if n < 10:
    return {"bin": bin_label, "n": n,
            "status": "insufficient_data"}

result = {
    "bin": bin_label,
    "n": n,
    "mean": float(np.mean(dT)),
    "median": float(np.median(dT)),
    "std": float(np.std(dT, ddof=1)),
    "skewness": float(stats.skew(dT)),
    "kurtosis": float(stats.kurtosis(dT)),
    "p_shapiro": float(stats.shapiro(dT[:500])[1]
                        if n >= 20 else np.nan),
}

# Bimodality coefficient: BC > 0.555 suggests bimodal
s = result["skewness"]
k = result["kurtosis"]
bc = (s**2 + 1) / (k + 3 * (n-1)**2 / ((n-2)*(n-3)))
result["bimodality_coeff"] = float(bc)
result["bimodal_flag"] = bc > 0.555

# Hartigan's dip test (simplified: compare IQR)
q25, q75 = np.percentile(dT, [25, 75])
iqr = q75 - q25
result["iqr"] = float(iqr)
result["iqr_ratio"] = float(iqr / result["std"]
                            if result["std"] > 0 else 0)

# RCMB key diagnostic: does deepest bin have
# larger variance than shallower bins?
result["rcmb_note"] = (
    "Sign-flip signature: look for increased variance, "
    "non-zero skewness, or bimodality_coeff > 0.555 "
    "in deepest delta-bin relative to shallower bins."
)
return result

```

Listing 7: RCMB Module 3: Episenter-Shift Test

```

"""rcmb_episenter.py -- Episenter shift analysis."""
import numpy as np

def episenter_shift_test(cat, dT_col='dT',
                        keys=('delta', 'R_eff', 'z')):
    """Rebin A_ISW by alternative episenters.

    RCMB Thesis 4: If delta is physically meaningful,
    shifting episenter should produce STRUCTURED
    redistribution, not random scatter.
    """
    results = {}

```

```

for key in keys:
    values = np.array(cat[key])
    dT = np.array(cat[dT_col])

    # Bin into quintiles
    edges = np.percentile(values[np.isfinite(values)],
                           [0, 20, 40, 60, 80, 100])

    bins = []
    for i in range(len(edges) - 1):
        mask = (values >= edges[i]) & (values < edges[i+1])
        if i == len(edges) - 2: # include right edge
            mask |= (values == edges[i+1])
        dT_bin = dT[mask & np.isfinite(dT)]

        if len(dT_bin) > 5:
            bins.append({
                "bin_center": float(
                    0.5*(edges[i]+edges[i+1])),
                "n": int(len(dT_bin)),
                "mean_dT": float(np.mean(dT_bin)),
                "std_dT": float(
                    np.std(dT_bin, ddof=1)),
            })

    # Spearman trend test across bins
    if len(bins) >= 3:
        from scipy.stats import spearmanr
        centers = [b["bin_center"] for b in bins]
        means = [b["mean_dT"] for b in bins]
        rho, p = spearmanr(centers, means)
        trend = {"spearman_rho": float(rho),
                  "p_value": float(p),
                  "structured": p < 0.05}
    else:
        trend = {"structured": False,
                  "note": "too few bins"}

    results[key] = {
        "bins": bins,
        "trend": trend
    }

# RCMB diagnosis: compare trends
results["rcmb_diagnosis"] = {
    "primary_episenter": "delta",
    "structured_alternatives": [
        k for k, v in results.items()
        if isinstance(v, dict) and
        v.get("trend", {}).get("structured", False)
    ],
    "interpretation": (
        "If ONLY delta shows structured trend: "
        "delta is physically controlling. "
        "If multiple keys show trend: "
        "confounding variable exists. "
        "If NO key shows trend: "

```

```

        "signal may be noise."
    )
}
return results

```

Listing 8: RCMB Module 4: LoS Contamination Test

```

"""rcmb_los.py -- Line-of-sight contamination test."""
import numpy as np

def los_contamination_test(cat, T_map, nside,
                           n_shifts=100, dz_max=0.05):
    """Test if ISW signal is local to void or
    contaminated by LoS projection.

    Two independent tests:
    A) z-shuffle: shift each void in redshift
    B) angular randomization: random sky positions
    If signal persists under either, it is not
    local to the void positions.
    """

    from stacker import stack_voids # base pipeline

    # Original signal
    original = stack_voids(cat, T_map, nside)
    A_orig = np.nanmean(original['dT'])

    # --- Test A: z-shuffle ---
    A_zshuf = []
    for _ in range(n_shifts):
        cat_s = cat.copy()
        dz = np.random.uniform(-dz_max, dz_max, len(cat))
        cat_s['z'] = np.clip(cat['z'] + dz, 0.01, 0.30)
        r = stack_voids(cat_s, T_map, nside)
        A_zshuf.append(np.nanmean(r['dT']))
    A_zshuf = np.array(A_zshuf)

    # --- Test B: angular randomization ---
    A_angrand = []
    for _ in range(n_shifts):
        cat_r = cat.copy()
        cat_r['RA'] = np.random.uniform(0, 360, len(cat))
        cat_r['DEC'] = np.degrees(np.arcsin(
            np.random.uniform(-1, 1, len(cat))))
        r = stack_voids(cat_r, T_map, nside)
        A_angrand.append(np.nanmean(r['dT']))
    A_angrand = np.array(A_angrand)

    def sn(A_orig, A_null):
        s = np.std(A_null)
        return float((A_orig - np.mean(A_null)) / s
                     if s > 0 else 0)

    return {
        "A_original": float(A_orig),
        "zshuf_SN": sn(A_orig, A_zshuf),
        "zshuf_is_local": bool(

```

```

    abs(sn(A_orig, A_zshuf)) > 2),
"angrand_SN": sn(A_orig, A_angrand),
"angrand_is_local": bool(
    abs(sn(A_orig, A_angrand)) > 2),
"rcmb_verdict": (
    "PASS" if (abs(sn(A_orig, A_zshuf)) > 2
               and abs(sn(A_orig, A_angrand)) > 2)
    else "FAIL: signal not localized"
)
}

```

10.1 RCMB Compliance Checklist

Before unblinding, the following must be verified:

Table 12: RCMB pre-unblinding checklist

#	Requirement	Status
1	Proxy chain logged for every void	☐
2	Distribution stats (skew, kurtosis, BC) per bin	☐
3	Episenter-shift test (δ , R_v , z , e) complete	☐
4	LoS contamination $S/N > 2$ (signal is local)	☐
5	Multi-void-finder consistency (VF vs. VIDE vs. REV)	☐
6	$D_A \pm 10\%$ sensitivity run complete	☐
7	All 4 CMB maps consistent within 1σ	☐

Rule: If items 1–4 are not checked, the physics verdict from Stage 6 is suspended. RCMB compliance is a *prerequisite* for a valid kill decision, not an optional add-on.

11 Expected RCMB Signatures

If EFC is correct and the sign-flip is real, the RCMB layer predicts specific diagnostic signatures:

Table 13: RCMB signatures of a real sign-flip

RCMB test	If sign-flip is real	If sign-flip is artifact
Distribution (R1)	Deepest bin: increased σ , skew $\neq 0$, $BC > 0.555$	All bins have similar σ , Gaussian
Episenter (R2)	δ -trend is structured; R_v and z trends are weaker or absent	Trend appears in R_v or z equally: confound
Void-finder (R3)	Sign-flip persists across VF/VIDE/REV	Sign-flip present only in one algorithm
LoS test (R4)	Signal disappears with z -shuffle (local)	Signal persists (projection artifact)

The combination of R1–R4 separates four scenarios:

1. **Real sign-flip:** R1 shows bimodality, R2 shows δ -only trend, R3 consistent, R4 local. Verdict: EFC supported.
2. **Measurement artifact:** R3 or R4 fails. Verdict: pipeline problem, not physics.
3. **No signal:** R1 shows Gaussian in all bins, R2 shows no trend anywhere. Verdict: EFC falsified at this sensitivity.
4. **Non-local signal:** R1 or Stage 6 suggests signal, but R4 fails (LoS contamination). Verdict: signal present in data but not attributable to target voids; no physical interpretation permitted.

12 References

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